Do Libraries Organize Knowledge the Way People Actually Think?

A Plain-Language Guide to

My Capstone Research Project

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Every time you search a library catalog,
a century-old filing system
built by professional catalogers
is quietly shaping what you find.

This research asks: *does that system actually match the way your brain works?*

See pages 11-12 for definitions of terms

The Big Picture: What Problem Does This Research Address?

Imagine someone visits a library and searches for books about "foundations." Are they interested in nonprofits or construction? The library's catalog needs to have already decided *which kind of "foundation"* each book belongs to—and it needs that answer to match the way this individual thought about the word when he or she typed it.

This deceptively simple challenge is important to library and information science, and it turns out to be surprisingly hard to get right.

Libraries Use a Filing System Called LCSH

The **Library of Congress Subject Headings (LCSH)** is the master vocabulary that most American libraries use to categorize their collections. It was created and is maintained by the Library of Congress, and it contains hundreds of thousands of official subject labels—things like *"Foundations," "Nonprofit Organizations,"* or *"Cognition in children."*

> **What LCSH Actually Does:** When a librarian catalogs a new book, they look up official LCSH terms and attach one or more of these approved labels to the record. When you search the catalog, those labels help the system retrieve the right items—even if the author used different words from your search terms.

LCSH was built through what catalogers call **literary warrant**: headings were added whenever enough published books on a topic existed to justify a new label. It grew organically over more than a century—pragmatically, not scientifically. Nobody sat down with a theory of human psychology and designed it from scratch.

The "Vocabulary Problem": When Libraries and People Disagree

In 1987, a landmark study discovered something embarrassing: if you ask different people to provide a search term for the same object, the chance that any two of them will use *exactly the same word* is only about 1 in 8. People use wildly different words for the same things.

The mismatch between the terms librarians choose and the terms users type is called the **vocabulary problem**. It is one of the oldest challenges in information retrieval, and it has not gone away.

But this project asks a *deeper* version of the question: even when the words match, do the *categories* match? Does the way LCSH slices up the meaning of a word align with the way the human brain naturally organizes that meaning?

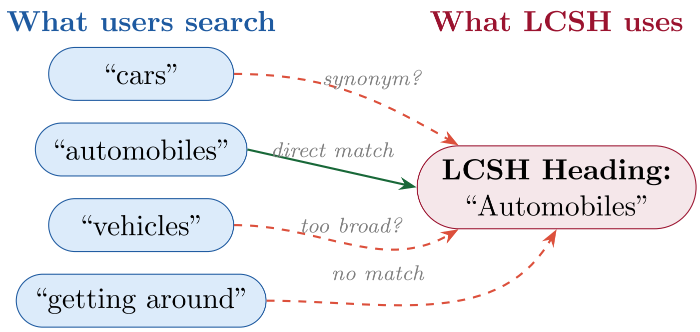

Figure 1: The vocabulary problem: many different user search terms compete to match a single LCSH heading. Only one user happens to use the exact official word. Dashed arrows show imperfect or missing matches.

Words With Multiple Meanings: Polysemy and Homonymy

Before diving into the experiments, it helps to understand a key concept: many words in English have more than one meaning, and linguists distinguish two flavors of this.

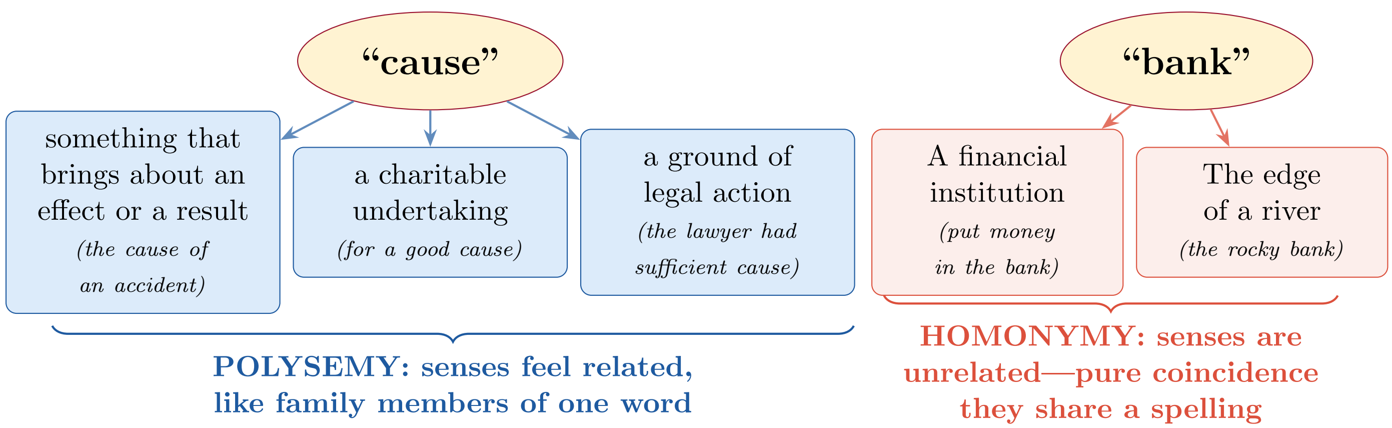

Figure 2: Two types of word ambiguity. Polysemous senses share a common origin and feel semantically connected; homonymous senses happen to share a spelling but are historically unrelated.

This distinction matters because LCSH must decide how to categorize each meaning of an ambiguous word. And the research asks: does the *type* of ambiguity (polysemy vs. homonymy) affect how well LCSH's categories align with how people actually process the word?

The Central Idea: Does LCSH Match How the Brain Thinks?

Modern brain and language research has revealed something surprising: meaning in the human mind is not organized like a tidy dictionary with clean, sharp-edged entries. Instead, words and their meanings exist in a kind of **semantic space**—a mental landscape where similar ideas are close together and different ideas are far apart.

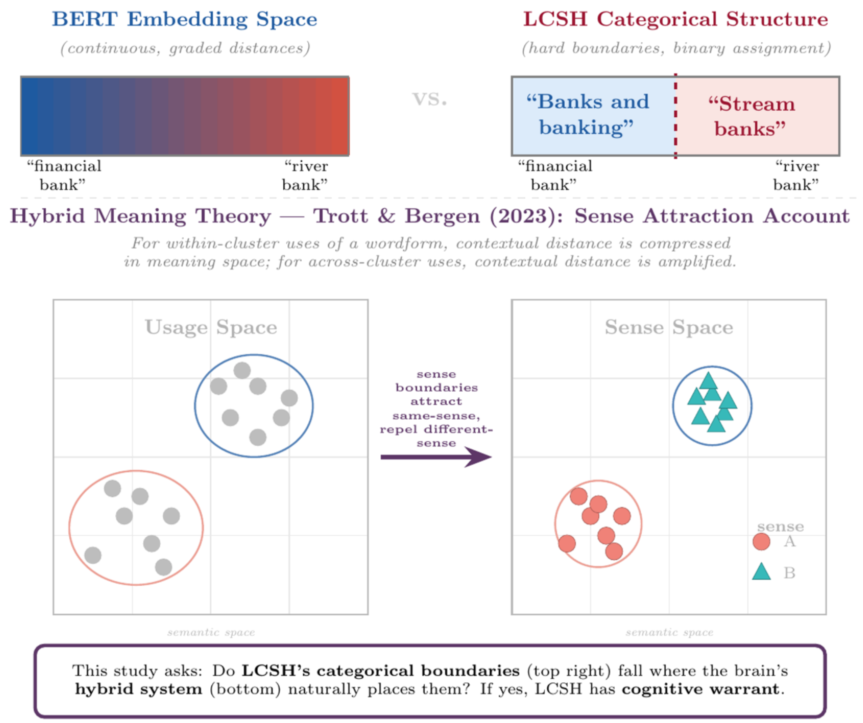

Figure 3: This figure presents the three representational systems the study works with. BERT gives us continuous, graded semantic distance. LCSH gives us hard categorical boundaries. And Trott and Bergen's hybrid meaning theory provides the behavioral framework that connects both to processing difficulty. These three are genuinely non-redundant.

The distance between two meanings of the same word in this mental landscape can be measured using an artificial intelligence tool called **BERT** (Bidirectional Encoder Representations from Transformers). BERT reads sentences and produces a numeric "location" for each word in context, allowing researchers to calculate how far apart two uses of a word actually are in meaning.

> **A Simple Analogy:** Think of meaning like a map of a city. "River bank" and "financial bank" might be in completely different neighborhoods—far apart on the map. But "bank loan" and "bank account" might be right next door to each other. BERT lets researchers calculate these distances numerically.

A key 2023 study by Trott and Bergen found that meaning appears to be **both** continuous (graded distances, like a map) **and** categorical (sharp boundaries, like neighborhoods). The brain uses *both* systems at once.

This project asks: **do the neighborhood boundaries in LCSH match the neighborhood boundaries in human minds?**

How LCSH Is Organized: A Hierarchy of Knowledge

LCSH does not just list subject headings—it organizes them into a **hierarchy** of broader and narrower terms. Think of it like a family tree for topics.

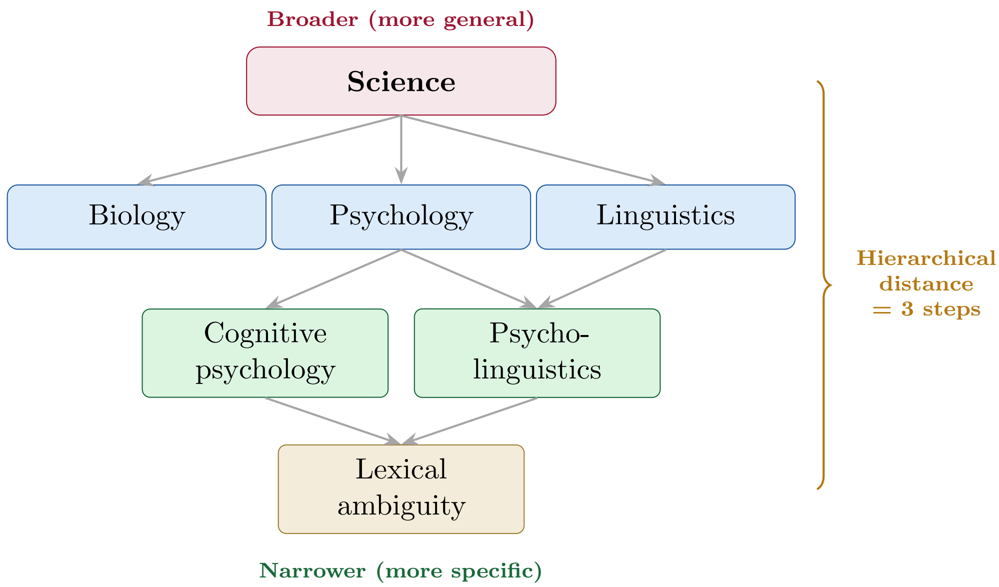

Figure 4: A simplified illustration of LCSH's tree-like hierarchy. Each heading sits inside broader parent headings. "Hierarchical distance" counts the number of steps between two headings—a key measurement in this research.

Why Does Hierarchy Matter? This research asks whether the *distance between two headings in the LCSH family tree* is cognitively meaningful. If "river bank" and "financial bank" are in completely different branches of the tree (far hierarchical distance), does the brain also treat them as more distinct? If LCSH's hierarchy tracks human cognition, it would validate the real-world value of all that careful organizational work by Library of Congress catalogers.

The Two Experiments

The research tests these ideas through two carefully designed experiments, each with 100–150 participants.

Experiment 1

"Does it make sense?"

Step 1: Participant sees two sentences using the same word consecutively in different contexts.

Step 2: They press a key as fast as possible to say whether each sentence "makes sense."

Measure: *How fast* did they respond? Slower = more mental effort needed.

Experiment 2

"Is this heading appropriate?"

Step 1: Participant reads a sentence about a topic (e.g., about a river bank).

Step 2: They rate how well various LCSH headings describe a document on that topic (1–7 scale).

Measure: *How high a rating* did each heading receive? Higher = better perceived fit.

Both experiments ask: Does crossing an LCSH category boundary—versus staying within one category—affect participant responses, *even after* accounting for how similar the two meanings already are according to BERT?

Figure 5: Overview of the two experiments. Experiment 1 measures unconscious processing speed; Experiment 2 measures conscious judgment of appropriateness. Together they test whether LCSH boundaries are cognitively meaningful both implicitly and explicitly.

A Concrete Example of What Participants See

Example trial for the word "palm":

Experiment 1 (sensibility judgment):
Prime sentence: "It was an introductory course."
Target sentence: "It was a charted course."
→ Participant presses YES or NO: does this make sense?
→ **Key question:** Are they faster responding to the target sentence when both sentences use "course" in the *same* LCSH category?

Experiment 2 (appropriateness rating):
Context: "The tourists relaxed under a row of tall palm trees."
Rate these LCSH headings for a document about this topic:
• "Palm trees" • "Tropical plants" • "Hand—Anatomy"
→ Participant rates each 1 (terrible fit) to 7 (perfect fit).

The Four Research Questions (Simply Put)

The research is organized around four questions, each building on the last:

1. **Independent effects:** Do *both* BERT-measured semantic distance *and* LCSH category membership independently affect how participants respond—or does only semantic distance matter?

builds on

2. **The ambiguity sweet spot:** Is LCSH's category boundary *most influential* when two meanings are neither very similar nor very different—right in the "could go either way" zone?

3. **Does LCSH structure add value?** Does the *distance between headings in LCSH's family tree* predict processing difficulty beyond BERT distance alone? And does polysemy vs. homonymy change the picture?

4. **Do both measures agree?** Do the unconscious processing patterns from Experiment 1 match the conscious ratings from Experiment 2, suggesting LCSH alignment is a stable, reliable property?

Figure 6: The four research questions, in plain language. Each question adds a new layer of complexity to understanding how LCSH aligns with human cognition.

How the Study Is Built: The Data and Word Selection

The research builds on an existing dataset of over 50,000 trials from 181 participants, originally collected by Trott and Bergen (2023). Those participants read pairs of sentences and judged whether they made sense, while their response times were recorded to the millisecond.

From that dataset, 25 words were carefully selected to represent a range of ambiguity types and meaning distances. Each word is then mapped to official LCSH headings by searching the Library of Congress's online database.

Example LCSH Mappings:

Word	Meaning 1 → LCSH Heading	Meaning 2 → LCSH Heading
bank	"Banks and banking"	"Riparian areas"
appeal	"Charisma (Personality trait)"	"Appellate procedure"
fan	"Fans (Machinery)"	"Fans (Persons)"
bat	"Bats"	"Baseball bats"

The key question for each word pair: are these two LCSH headings in the same branch of the hierarchy (**same category**) or different branches (**different category**)? And does that distinction predict how fast and accurately people respond?

What Would Confirm or Challenge the Theory?

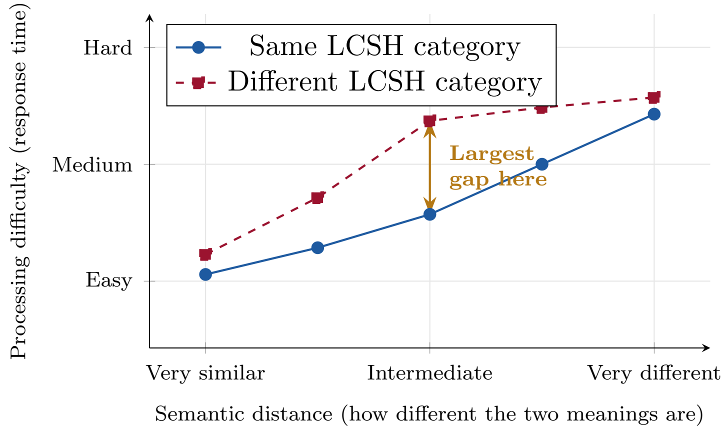

Figure 7: Predicted pattern if the research hypotheses are confirmed. The gap between same-category and different-category word pairs should be *largest* at intermediate semantic distances—the "could go either way" zone. At very similar or very different meanings, the LCSH boundary matters less.

If LCSH boundaries *are* cognitively valid: Participants should be noticeably slower (more mental effort) when processing sentences that cross LCSH category boundaries, compared to sentences that stay within the same LCSH category—especially when the meanings are moderately different from each other.

If LCSH boundaries *are not* cognitively valid: Crossing an LCSH boundary would have no measurable effect on processing speed or appropriateness ratings, suggesting the categories are organizationally convenient but mentally invisible.

Possible Implications: Why Should Anyone Care?

The results of this study—whichever direction they go—carry practical consequences for how libraries are built and how people find information.

If LCSH Boundaries Align With Cognition

Good news for libraries and catalogers: A century of pragmatic, expert cataloging judgment would be shown to have produced a system that genuinely mirrors how people process meaning—even though it was never designed with cognitive science in mind.

- **Justification for maintaining LCSH:** Provides the first empirical evidence that LCSH's subject boundaries serve a real cognitive function, countering arguments that subject headings are obsolete in an era of keyword search.
- **Support for hierarchical structure:** If hierarchical distance predicts cognitive difficulty, this validates the Library of Congress's careful work building and maintaining the LCSH tree structure—and supports keeping it in modern digital library systems.
- **Guidance for AI systems:** AI-powered search and recommendation systems could be improved by incorporating LCSH categorical structure, since it would be shown to carry cognitive information that pure text-distance measures miss.

If LCSH Boundaries Do *Not* Align With Cognition

Actionable intelligence for reform: A mismatch would not mean LCSH is useless—it would point precisely to which types of boundaries need revision and why.

- **Targeted vocabulary revision:** The study would identify exactly which words and categories show the greatest misalignment, giving vocabulary editors a data-driven priority list for updates.
- **Polysemy vs. homonymy:** If homonymous boundaries (bat the animal vs. bat as sports equipment) align with cognition but polysemous boundaries (related senses of "spring") do not, LCSH editors would know to focus revision energy on polysemous terms.
- **Discovery system design:** Library software developers could use these findings to build smarter disambiguation prompts—for example, asking users "Did you mean the river bank or the financial institution?" at precisely the moments when the distinction is most cognitively ambiguous.

Broader Implications Beyond LCSH

A method for evaluating any controlled vocabulary: The appropriateness rating paradigm developed in Experiment 2 is entirely general. It can be applied to evaluate MeSH (the medical subject headings used in PubMed), legal classification systems, corporate taxonomies, and any other knowledge organization system—producing actionable data to inform revision decisions.

For AI and search engines: Search engines like Google and Bing use statistical methods to understand meaning, but they typically do not use categorical subject structures like LCSH. If LCSH boundaries are shown to carry independent cognitive information, this would make a strong case for hybrid systems that combine both approaches.

For cognitive science: The study contributes to a fundamental debate about how the brain organizes language: Is meaning fundamentally continuous (like a smooth spectrum) or fundamentally discrete (like separate boxes)? The evidence points toward "both"—but this study tests that theory against professionally imposed, culturally significant boundaries, not just experimental laboratory categories.

The February 2026 LC policy change: Interestingly, the Library of Congress recently made a technical change—moving form subdivisions out of MARC records into a separate vocabulary (LCGFT). This was done for technical data-management reasons, but this research could provide the first cognitive validity evidence for or against such restructuring decisions, turning future policy choices from debates into data-driven conclusions.

Project Timeline at a Glance

Figure 8: Simplified project timeline. Shaded phases are complete or underway; lighter phases are planned for Spring Quarter 2026.

Summary: The One-Paragraph Version

My capstone asks: When libraries use their official subject heading system (LCSH) to categorize books and articles, do those categories match the way the human brain naturally organizes meaning? To answer this, I am running two experiments in which people respond to pairs of sentences containing ambiguous words, and I measure how their unconscious response times and conscious ratings correspond to LCSH's formal category boundaries. I use an artificial intelligence language model (BERT) to measure how *similar* two meanings of the same word actually are, and then test whether LCSH boundaries predict people's responses *over and above* that similarity—especially in the murky middle zone where meaning is most ambiguous. The results will tell us whether a century of professional library cataloging has—perhaps accidentally—produced a system that truly reflects human cognitive structure, or whether there are specific places where it falls short and could be improved.

Definitions of Terms

Ambiguity type: The classification of an ambiguous word as a polyseme (multiple, related meanings) or a homonym (multiple, unrelated meanings).

BERT (Bidirectional Encoder Representations from Transformers): A large language model that generates word embeddings capturing meaning-in-context rather than static word meanings. These vectors or "locations" in high-dimensional space enable measurement of continuous semantic dissimilarity independently of categorical sense assignments.

BERT cosine distance: Quantitative measure of semantic distance (i.e., dissimilarity) between word senses, operationalized in this study as the cosine distance between BERT-derived contextualized word embeddings. Cosine distance can be defined as $(1-(\text{cosine similarity}))$, where cosine similarity has a range of $[-1, 1]$. The range of cosine distance is thus $[0, 2]$, where two proportional vectors have a cosine distance of 0, orthogonal vectors have a distance of 1, and opposite vectors have a distance of 2. As shown in Figure 9, cosine distance values of word pairs range from 0 (identical meanings) to 1 (no relatedness) to 2 (opposite meanings).

Cognitive warrant: The principle that classification structures should align with empirically-measured patterns of human cognitive processing. This study establishes cognitive warrant through response time and appropriateness rating data.

Discovery system: Integrated library search platforms (e.g., Ex Libris Primo) that provide unified keyword searching across multiple databases while incorporating controlled vocabulary through relevance boosting and faceted filtering. Distinguished from traditional OPACs by their Google-like interfaces and behind-the-scenes controlled vocabulary integration.

Hierarchical distance: The combined shortest path length from two LCSH subject headings to their lowest common ancestor within the LCSH polyhierarchical graph.

Homonymy: A single word form possessing multiple *unrelated* meanings that do not share conceptual features (e.g., "bank" as a financial institution vs. the side of a river). In lexicographic practice, homonymous senses are typically listed as separate headword entries. Homonymous words are called homonyms.

Hybrid meaning theory: Trott and Bergen's (2023) theoretical framework positing that word senses are both continuously graded (processing difficulty proportional to semantic distance) and categorically distinct (producing processing discontinuities at sense boundaries).

LCSH (Library of Congress Subject Headings): The controlled vocabulary developed by the Library of Congress for subject cataloging in bibliographic records. Provides pre-coordinated subject strings with topical, geographic, chronological, and form subdivisions.

Literary warrant: The principle that classification structures should reflect what exists in published literature—new headings are added when sufficient publications on a topic accumulate. Contrasts with cognitive warrant.

Parenthetical qualifier: Disambiguating text in parentheses added to LCSH headings to distinguish polysemes/homonyms or clarify the meaning of the referent (e.g., "Java

(Indonesia)" vs. "Java (Computer program language)").

Polysemy: A single word form possessing multiple *related* meanings (e.g., "paper" as a material, document, assignment, or newspaper). In lexicographic practice, polysemous senses are typically listed as definitions under the same headword entry. Polysemous words are called polysemes.

Pre-coordination: The practice of combining multiple concepts into single subject heading strings (e.g., "Libraries–Academic–United States") before documents are indexed. Contrasts with post-coordination where users combine terms at the time of search.

Processing difficulty or cognitive load: The computational demands placed on cognitive resources during language comprehension, operationalized in this study as response times.

Scope note: Explanatory text in LCSH authority records clarifying the intended usage of a heading and distinguishing it from related terms. Enables catalogers to apply headings consistently and helps users to determine what type of material may be found under that heading.

Sense boundary: The categorical distinction between different meanings of a polysemous or homonymous word. In dictionaries, sense boundaries for homonyms can be marked by numbered headwords (e.g., bank^1 and bank^2). Senses for polysemes are usually listed as definitions under the same headword. In LCSH, boundaries are marked by separate authority records, parenthetical qualifiers, or scope notes.

User warrant: The principle that classification should reflect actual user search terminology and behavior. Related to but distinct from cognitive warrant—users may search with terms they find cognitively "easy" even when professional alternatives would retrieve better results.

Word meaning and senses: Although word meaning can refer to the full semantic range associated with a lexical form, while senses are the various interpretations that emerge in different contexts, here they are used interchangeably.

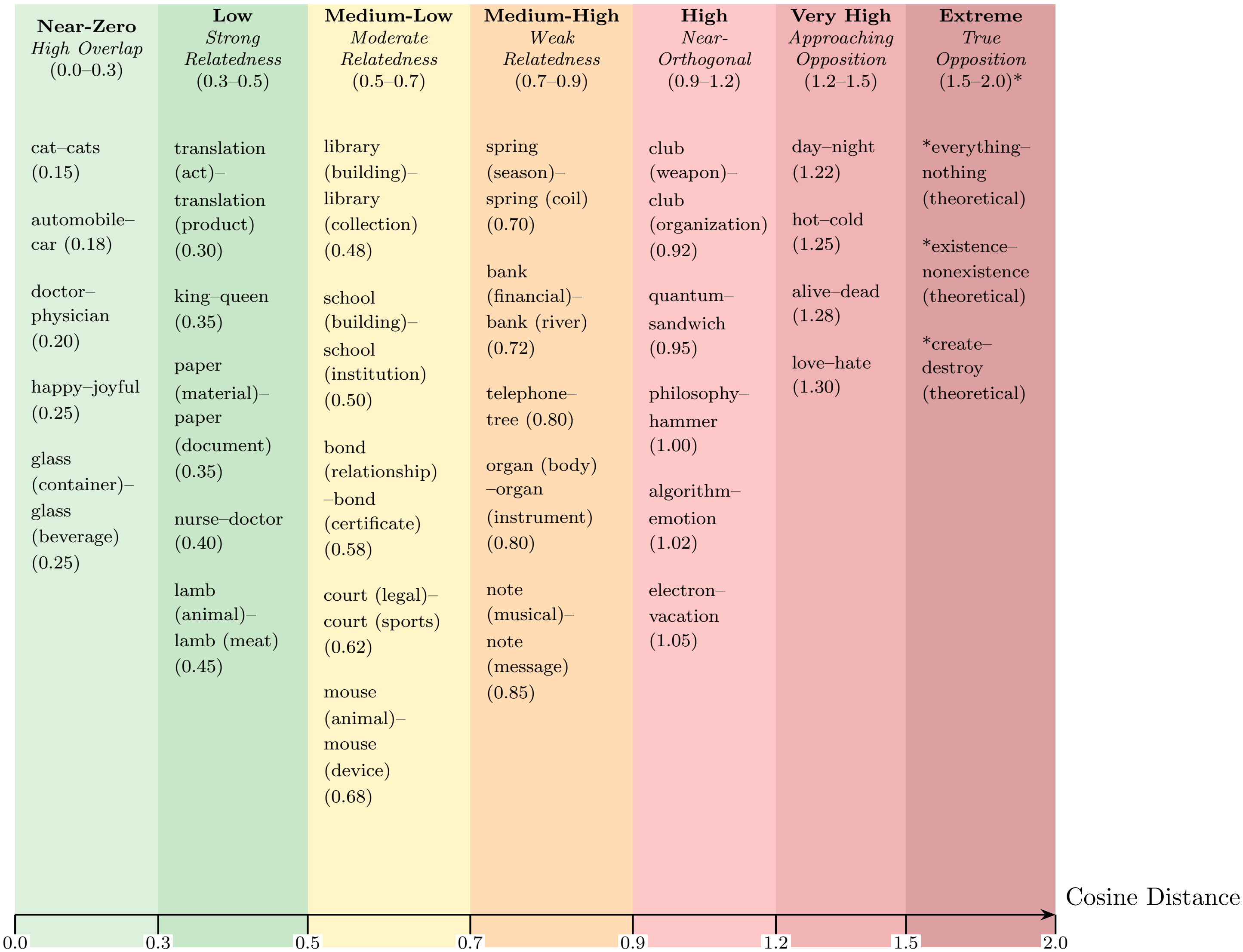

Figure 9: Semantic distance continuum showing examples of word pairs across the full range of cosine distance values (0–2). The concentration of polysemous examples in the 0.4–0.8 range demonstrates the "ambiguous" zone where categorical boundaries become critical for disambiguation. *Asterisks in the Extreme region indicate theoretical examples; values near 2.0 are extremely rare in practice with current word embeddings such as BERT and ELMo. The largest BERT cosine distance in Trott and Bergen stimuli is 0.7499.